THE FORMAL COMPLETION OF AN AFFINE SCHEME ALONG A CLOSED SUBSET, AND ITS TYPES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays one colimit description of the completion functor; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- A closed subset with quasi-compact complement in an affine scheme is cut out by a finitely generated ideal. Algebra, Lemma algebra-lemma-qc-open, invoked at proof lines 3289-3291. Produces the finitely generated ideal in the first case.
- The completion of a Noetherian ring is Noetherian. Algebra, Lemma algebra-lemma-completion-Noetherian-Noetherian, invoked at proof lines 3293-3295. Gives the third assertion once the second is known.
- The formal completion of an affine scheme as a functor, and the description of morphisms into a formal spectrum. Same chapter, the definition of the completion functor in force in the statement, and the description of morphisms into a formal spectrum. Fixes the functor whose identification with the formal spectrum of the completion is proved.

Source reference. Original lemma 0316: sources/stacks-chow/tags.txt, source lines 3973–3973. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 00F6: sources/stacks-chow/tags.txt, source lines 560–560. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $X = \mathrm{Spec}(A)$ be an affine scheme over S. Let $T \subset X$ be a closed subset. Let $X_{/T}$ be the formal completion of X along T.*

(1) *If $X \setminus T$ is quasi-compact, i.e., T is constructible, then $X_{/T}$ is adic*.*

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(2) *If $T = V(I)$ for some finitely generated ideal $I \subset A$, then $X_{/T} = Spf(A^\wedge)$ where $A^\wedge$ is the I-adic completion of A.*
(3) *If X is Noetherian, then $X_{/T}$ is Noetherian.*

Proof. By Algebra, Lemma 00F6 if (1) holds, then we can find an ideal $I \subset A$ as in (2). If (3) holds then we can find an ideal $I \subset A$ as in (2). Moreover, completions of Noetherian rings are Noetherian by Algebra, Lemma 0316. All in all we see that it suffices to prove (2).

Proof of (2). Let $I = (f_1, \ldots, f_r) \subset A$ cut out T. If $Z = \mathrm{Spec}(B)$ is an affine scheme and $g : Z \to X$ is a morphism with $g(Z) \subset T$ (set theoretically), then $g^\sharp(f_i)$ is nilpotent in B for each i. Thus I^n maps to zero in B for some n. Hence we see that $X_{/T} = \mathrm{colim}\, \mathrm{Spec}(A/I^n) = \mathrm{Spf}(A^\wedge)$. □